Lecture 26

Continuous integrands and the fundamental theorem

Uses.

Summary 2: The Archimedean property supplies arbitrarily fine partitions.
Summary 5: Function limits obey algebraic and order laws. Continuity is relative to the domain, preserves sequence limits and is closed under products and quotients with a nonzero denominator.
Summary 6: A continuous image of a real interval is an interval.
Summary 7: Continuous functions on $[a, b]$ are bounded and attain both extreme values. A derivative is a difference-quotient limit, and differentiability implies continuity.
Summary 8: Sum, product and chain rules hold. For positive integers n, $(x^n)' = nx^{n-1}$.
Summary 9: A continuous function on an interval whose derivative is zero throughout the interior is constant.
Summary 25: Heine–Cantor gives uniform continuity on a closed finite interval. Darboux gaps test integrability. Integral linearity, order, additivity and the absolute-value bound hold. Tagged-sum errors are bounded by the gap. Monotone real functions on $[a, b]$ are integrable.

Objectives. *On a closed finite interval, uniform continuity makes continuous functions integrable. We then study an integral whose upper endpoint moves. Its derivative recovers the integrand at interior continuity points. This gives the endpoint formula for a definite integral and proves the definite substitution and parts rules.*

By the end of this lecture you will be able to:

- Prove integrability of continuous functions and obtain an integral mean value.
- Prove continuity and differentiate an accumulation integral at an interior continuity point of its integrand.
- Evaluate a definite integral from an antiderivative with its endpoint conditions.
- Apply definite substitution and integration by parts with their hypotheses.

Throughout, $a < b$ are real numbers. A function on $[a, b]$ is continuous at an endpoint in the relative-domain sense defined in Lecture 5 and recalled in Lecture 25.

26.1 Continuous functions are integrable

Theorem 26.1 (Continuous integrability). *Every continuous real function on $[a, b]$ is Riemann integrable.*

Proof. The extreme value theorem, Theorem 7.1, bounds f on $[a, b]$. Given $\varepsilon > 0$, uniform continuity supplies $\delta > 0$ such that points less than δ apart have values less than $\eta = \varepsilon/(2(b-a))$ apart. Consider any partition with mesh below δ. Such partitions exist by the Archimedean property from Lecture 2. In each closed subinterval, every pair x, y satisfies $f(x) \le f(y) + \eta$. Taking the supremum over x and the infimum over y gives $M_i - m_i \le \eta$, without assuming either bound is attained. Consequently,

$$U(P, f) - L(P, f) \le \sum_i \eta \Delta x_i = \eta(b-a) = \frac{\varepsilon}{2} < \varepsilon.$$

The Darboux criterion proves integrability. The estimate holds for every partition of sufficiently small mesh, so Lecture 25's tagged-sum bound also applies uniformly over their tags. ∎

Uniform continuity supplies one mesh size for the whole interval. Continuity at separate points alone would give different local sizes. Heine–Cantor is the proved step that permits their replacement by one choice. No differentiation theorem was needed for this conclusion.

26.2 Finitely many discontinuities

A *discontinuity* is a domain point at which a function is not continuous. The following result is recorded with a proof sketch [S].

Theorem 26.2 (Finite-discontinuity integrability; sketch status). *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Riemann integrable.*

Proof sketch. Choose $M > 0$ with $|f| \le M$. Fix $\varepsilon > 0$. Cover the finitely many discontinuities by open intervals, truncated at a and b, whose total length is below $\varepsilon/(4M)$. Choose their interior boundary points outside the finite set of discontinuities. Endpoint neighbourhoods may be one-sided. Merge overlapping intervals.

Ignore empty or zero-length complementary pieces. If there are none, only the covered-cell estimate below is needed. Each remaining closed piece contains no discontinuities. The restriction of f to each piece is continuous and hence uniformly continuous by Heine–Cantor. There are only finitely many pieces, so one positive mesh choice bounds their subinterval oscillations by $\eta = \varepsilon/(2(b-a))$. Refine separately on these pieces and include all covering-interval boundary points in the partition. The covered cells contribute less than $2M \cdot \varepsilon/(4M) = \varepsilon/2$ to the Darboux gap. The other cells contribute at most $\eta(b-a) = \varepsilon/2$. The total gap is less than ε, proving integrability by the criterion. If there are no discontinuities, use the continuous case. ∎

26.3 A mean value for an integral

Theorem 26.3 (Integral mean value). *If $f : [a, b] \to \mathbb{R}$ is continuous and $a < b$, then some $c \in [a, b]$ satisfies*

$$\int_a^b f(x)\,\mathrm{d}x = f(c)(b - a).$$

Proof. The extreme value theorem, Theorem 7.1, gives points attaining the minimum m and maximum M. Order and the constant integral give

$$m(b - a) \le \int_a^b f(x)\,\mathrm{d}x \le M(b - a).$$

Since $b - a > 0$, the average $\mu = (b-a)^{-1}\int_a^b f$ lies in $[m, M]$. The interval-image consequence of Theorem 6.4 supplies $c \in [a, b]$ with $f(c) = \mu$. Multiplying by $b - a$ proves the formula. ∎

For example, Lecture 25 computed $\int_0^1 x^2\,\mathrm{d}x = 1/3$. The mean-value equation is therefore $c^2 = 1/3$ with $c \in [0, 1]$. It is satisfied by $c = 1/\sqrt{3}$. This is the point where the function's value equals its integral average.

26.4 An integral with a moving endpoint

For an integrable function we extend the endpoint convention by

$$\int_u^u f(t)\,\mathrm{d}t = 0, \qquad \int_v^u f(t)\,\mathrm{d}t = -\int_u^v f(t)\,\mathrm{d}t \quad (u < v).$$

Additivity then holds for any order of endpoints in the original interval. For three distinct points, order them increasingly, use adjacent-interval additivity, and rearrange by the reversal convention. If two points coincide, the zero integral gives the same identity.

If $|f| \le M$ and $u < v$, order gives $-M(v - u) \le \int_u^v f \le M(v - u)$. Reversal handles $v < u$. Thus, for either endpoint order,

$$\left|\int_u^v f(t)\,\mathrm{d}t\right| \le M|v - u|. \tag{26.1}$$

Here f is assumed integrable on the interval between the endpoints.

Theorem 26.4 (Fundamental theorem: differentiating an integral). *Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable, with $a < b$, and define*

$$A(x) = \int_a^x f(t)\,\mathrm{d}t \qquad (x \in [a, b]).$$

Then A is continuous on $[a, b]$. At every interior point x where f is continuous, A is differentiable and $A'(x) = f(x)$.

Proof. Restrictions of an integrable function to subintervals are integrable by Lecture 25. At the degenerate endpoint we use the zero convention. Choose $M > 0$ with $|f| \le M$. Additivity and Eq. (26.1) give

$$|A(y) - A(x)| = \left|\int_x^y f(t)\,\mathrm{d}t\right| \le M|y - x|.$$

For a prescribed error $\varepsilon > 0$, the choice $\delta = \varepsilon/M$ proves continuity, including at the endpoints.

Now let $x \in (a, b)$ be a continuity point of f. For $\varepsilon > 0$, choose $\delta > 0$ so that $|f(t) - f(x)| < \varepsilon/2$ whenever $t \in [a, b]$ and $|t - x| < \delta$. Take $h \neq 0$ with $|h| < \delta$ and $x + h \in [a, b]$. Every t between x and $x + h$ satisfies that bound. Linearity and additivity give

$$A(x + h) - A(x) - hf(x) = \int_x^{x+h} \big(f(t) - f(x)\big)\, \mathrm{d}t.$$

The constant integral equals $hf(x)$ for either sign of h. Applying Eq. (26.1) to the error integrand gives

$$\left|\frac{A(x + h) - A(x)}{h} - f(x)\right| \leq \frac{(\varepsilon/2)|h|}{|h|} = \frac{\varepsilon}{2} < \varepsilon.$$

Thus the difference quotient tends to $f(x)$, which proves the derivative formula. The argument covers positive and negative increments. ∎

Figure 26.1 illustrates the error term in the proof. The integral over a short interval differs from a rectangle of height $f(x)$. Continuity makes the error small compared with the interval length.

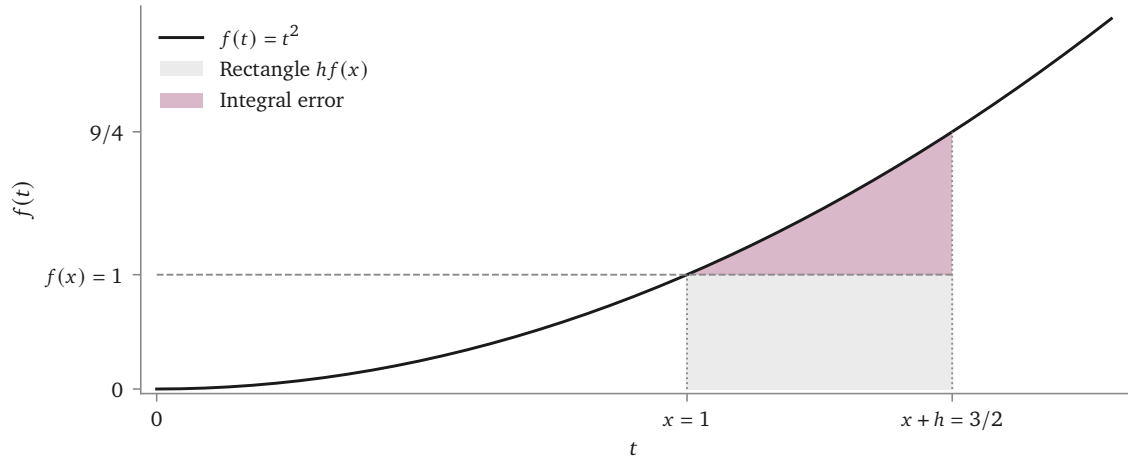


Figure 26.1: For $f(t) = t^2$, $x = 1$ and $h = 1/2$, the orange cap is $\int_x^{x+h} (f(t) - f(x))\, \mathrm{d}t$. The grey rectangle has area $hf(x)$.

Example 26.5 (Differentiating without evaluating the integral)**.** For $x \in [-2, 2]$, define

$$B(x) = \int_{-2}^{x} \sqrt{1 + t^2}\, \mathrm{d}t.$$

To apply the fundamental theorem, we check the integrand's continuity. Fix $c \in \mathbb{R}$. Rationalising the difference gives

$$\left|\sqrt{1 + t^2} - \sqrt{1 + c^2}\right| = \frac{|t - c||t + c|}{\sqrt{1 + t^2} + \sqrt{1 + c^2}} \leq \frac{|t - c||t + c|}{2}.$$

If $|t - c| < 1$, then $|t + c| \leq 2|c| + 1$. For any $\varepsilon > 0$, taking $|t - c| < \min\{1, 2\varepsilon/(2|c| + 1)\}$ makes the displayed error less than ε. The integrand is therefore continuous. The fundamental theorem now gives

$$B'(x) = \sqrt{1 + x^2} \quad (-2 < x < 2), \qquad B'(0) = 1.$$

This derivative was obtained directly from the integral definition.

Example 26.6 (A moving endpoint and the chain rule)**.** Let $F(u) = \int_0^u e^{-t^2}\,\mathrm{d}t$ and $G(x) = F(x^2)$. The integrand is continuous, so the fundamental theorem gives $F'(u) = e^{-u^2}$. The chain rule therefore gives

$$G'(x) = F'(x^2)\,2x = 2xe^{-x^4}.$$

The variable t is local to the integral; the endpoint x^2 supplies the additional factor $2x$.

Now let both endpoints move:

$$H(x) = \int_x^{x^2} e^{-t^2}\,\mathrm{d}t = F(x^2) - F(x).$$

Finite additivity and the reversal convention justify the difference for every real x, including $0 < x < 1$, where $x^2 < x$. Differentiating each term gives

$$H'(x) = 2xe^{-x^4} - e^{-x^2}, \qquad H'(0) = -1.$$

The upper endpoint contributes its derivative times the integrand there; the lower endpoint contributes the corresponding term with a minus sign.

Example 26.7 (Continuity does not guarantee a derivative)**.** Let f on $[-1, 1]$ be zero for $t \le 0$ and one for $t > 0$. It is monotone, so it is integrable by Lecture 25. Define $A(x) = \int_{-1}^{x} f(t)\,\mathrm{d}t$. For $x \le 0$ this is a zero integral. For $x > 0$, partition $[0, x]$ into n equal cells. Its lower sum is $x - x/n$ and its upper sum is x: only the first cell has infimum zero. Passing to the limit gives $\int_0^x f = x$. Hence

$$A(x) = \begin{cases} 0, & -1 \le x \le 0, \\ x, & 0 < x \le 1. \end{cases}$$

The function A is continuous at zero. Its difference quotient there is zero for negative increments and one for positive increments, so it has no derivative there. The integrand is discontinuous at precisely that point. The fundamental theorem did not assert a derivative there.

26.5 Evaluation by an antiderivative

An *antiderivative* of f on an open interval is a differentiable function F satisfying $F' = f$ there. When we use one on $[a, b]$, we require F to be continuous at both endpoints and $F' = f$ on (a, b). For a continuous f on an open interval J, choose any base point $c \in J$ and define $F(x) = \int_c^x f(t)\,\mathrm{d}t$ for $x \in J$. Each finite integral exists by Theorem 26.1. Additivity gives $F(x + h) - F(x) = \int_x^{x+h} f(t)\,\mathrm{d}t$. The same difference-quotient proof as in Theorem 26.4 therefore gives $F'(x) = f(x)$ at every $x \in J$, including the base point. Thus continuous functions have antiderivatives even when no elementary formula has been found.

Theorem 26.8 (Newton–Leibniz formula)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous, where $a < b$. If F is continuous on $[a, b]$ and differentiable on (a, b) with $F' = f$, then*

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Proof. The function f is integrable by Theorem 26.1. Let $A(x) = \int_a^x f$. The first fundamental theorem gives $A' = f$ on (a, b) and continuity on $[a, b]$. Consequently $H = F - A$ is continuous on $[a, b]$ and has derivative zero on (a, b). Theorem 9.4 gives $H(b) = H(a)$. Since $A(a) = 0$ and $A(b) = \int_a^b f$, rearranging this equality proves the formula. ∎

The notation $[F(x)]_a^b$ means $F(b) - F(a)$. For example, the power rule gives $(x^4/4)' = x^3$ on $\mathbb{R}$. Both functions are continuous at 0 and 2, so

$$\int_0^2 x^3 \, \mathrm{d}x = \left[\frac{x^4}{4}\right]_0^2 = 4.$$

The existence proof and the evaluation step have different roles: continuity proves that the integral exists, and the antiderivative computes its value.

26.6 Definite substitution and integration by parts

For these rules, all differentiable functions mentioned are defined on open intervals containing the relevant closed intervals. Their derivatives are required to be continuous on the closed intervals where they are used. This states the endpoint regularity needed by the integral formulas.

Theorem 26.9 (Definite substitution). *Let $f : J \to \mathbb{R}$ be continuous on an open interval J. Let g be differentiable on an open interval containing $[a, b]$, with g' continuous on $[a, b]$ and $g([a, b]) \subseteq J$. Then*

$$\int_a^b f(g(x))g'(x) \, \mathrm{d}x = \int_{g(a)}^{g(b)} f(u) \, \mathrm{d}u.$$

The endpoint integral uses the reversal and zero conventions when needed.

Proof. Choose an antiderivative F of f on J, whose existence was just proved. The composition $F \circ g$ is continuous on $[a, b]$ and the chain rule gives $(F \circ g)' = f(g)g'$ in the interior. The integrand is continuous: continuity of g puts its values near $g(c)$ when x is near c, and continuity of f then controls $f(g(x))$. Multiplication by the continuous g' preserves continuity. The evaluation theorem therefore gives the left side as $F(g(b)) - F(g(a))$. Applying the same theorem to f between $g(a)$ and $g(b)$ gives the right side. Reversal handles descending endpoints; coincident endpoints give zero on both sides. ∎

For example, take $g(x) = x^2 + 1$ on $[0, 1]$ and $f(u) = u^2$ on $\mathbb{R}$. The endpoint values are $g(0) = 1$ and $g(1) = 2$, and $g'(x) = 2x$. Hence

$$\int_0^1 2x(x^2 + 1)^2 \, \mathrm{d}x = \int_1^2 u^2 \, \mathrm{d}u = \left[\frac{u^3}{3}\right]_1^2 = \frac{7}{3}.$$

Monotonicity of g is not a hypothesis of this signed formula. For instance, $g(x) = x^2$ on $[-1, 1]$ with $f(u) = 1$ gives $\int_{-1}^1 2x \, \mathrm{d}x = \int_1^1 1 \, \mathrm{d}u = 0$. The factor g' retains its sign.

Theorem 26.10 (Definite integration by parts). *If u and v are differentiable on an open interval containing $[a, b]$ and both derivatives are continuous on $[a, b]$, then*

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

Proof. The product rule gives $(uv)' = u'v + uv'$. All three functions in this identity are continuous on $[a, b]$. The evaluation theorem and linearity give

$$[uv]_a^b = \int_a^b (u'v + uv') \, \mathrm{d}x = \int_a^b u'v \, \mathrm{d}x + \int_a^b uv' \, \mathrm{d}x.$$

Subtracting the first integral proves the formula. ∎

For example, in $\int_0^1 x(1+x)^4\,\mathrm{d}x$, choose $u(x) = x$ and $v(x) = (1+x)^5/5$. Then $u' = 1$ and $v' = (1+x)^4$. These polynomials satisfy all continuity requirements. Therefore

$$\int_0^1 x(1+x)^4\,\mathrm{d}x = \left[\frac{x(1+x)^5}{5}\right]_0^1 - \int_0^1 \frac{(1+x)^5}{5}\,\mathrm{d}x = \frac{32}{5} - \left[\frac{(1+x)^6}{30}\right]_0^1 = \frac{43}{10}.$$

Sheet 26 proves the even/odd symmetry rules and differentiates an integral with two moving endpoints. It also uses parts on an integral-defined function, where a closed formula for that function is unnecessary.

Summary 26.

- Theorems 26.1 and 26.2: continuous functions on $[a, b]$ are integrable [P]; bounded functions with finitely many discontinuities are integrable [S]. Here $a < b$.
- Theorem 26.3: a continuous integrand has an attained integral average $f(c) = (b-a)^{-1}\int_a^b f$ for some $c \in [a, b]$.
- Theorem 26.4: for an integrable f, its accumulation integral is continuous and has derivative $f(x)$ at every interior continuity point.
- Theorems 26.8 to 26.10: continuity and an antiderivative give endpoint evaluation. The definite substitution and parts rules follow from the chain and product rules with the stated continuity hypotheses.

Lecture 27 applies endpoint evaluation before taking limits to define improper integrals and functions given by such integrals.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, §§7.2–7.3.
S. R. Ghorpade and B. V. Limaye, *A Course in Calculus and Real Analysis*, 2nd ed., Springer, 2018, §§6.1–6.3.